# Component Packages

## Resistors, Capacitors, Inductors & IC Packages

How to read a datasheet — and pick the right footprint

Audience: Beginner → Intermediate Engineers

Through-hole • SMD • Leadless • BGA • Selection criteria • Assembly

# Why Component Packages Matter

Six dimensions in every package decision

## Why Component Packages Matter

### Footprint
Determines PCB landpattern and placement density

### Assembly
Hand-solder vs reflow, pick-and-place feasibility

### Thermal
How heat leaves the die — lead frame, pad, or balls

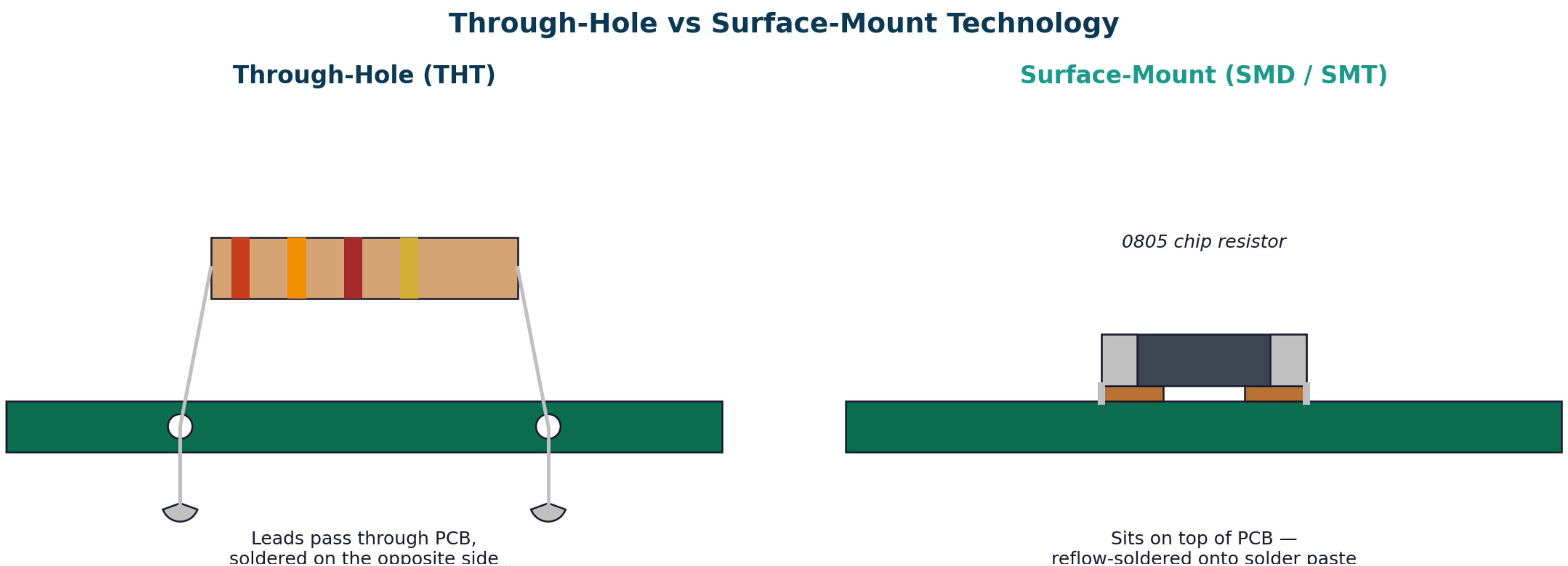
### Electrical
Lead inductance, ESR/ESL, parasitic capacitance

### Cost
Larger = cheaper to assemble, smaller = cheaper to ship

### Reliability
Solder-joint stress, vibration, thermal-cycle endurance

# Through-Hole vs Surface-Mount

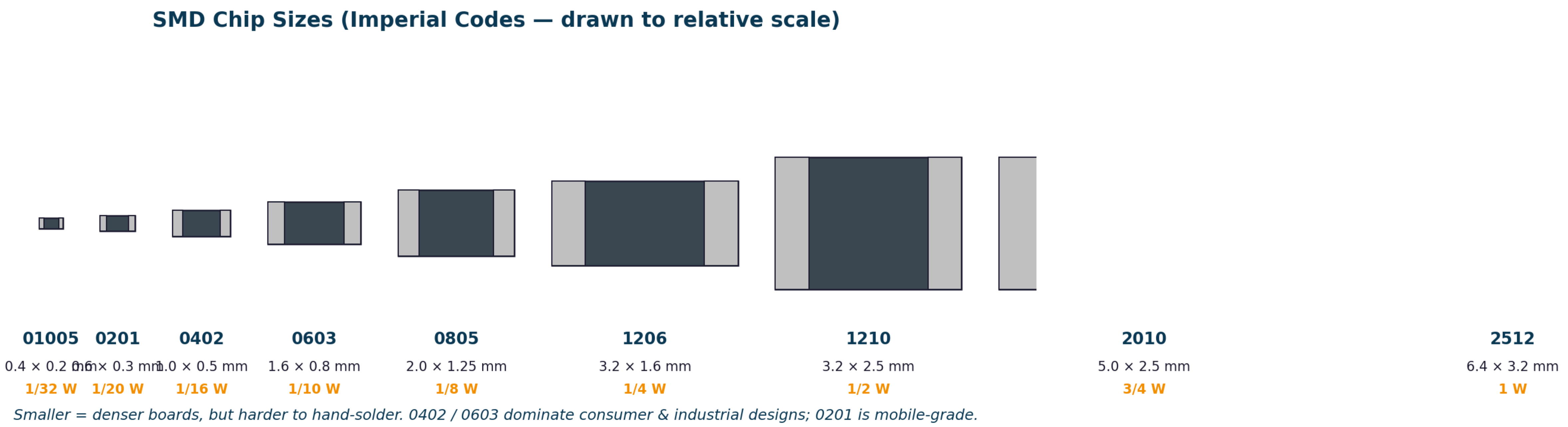
Two assembly worlds — and where each still wins




| | Through-Hole (THT) | Surface-Mount (SMD) |
|---|---|---|
| **Mechanical strength** | Excellent (lead anchors) | Good (solder only) |
| **Assembly** | Wave / hand soldering | Reflow oven, automated SMT |
| **Density** | Low | Very high (both sides) |
| **Best for** | Connectors, big power, prototyping | Most production: passives, ICs, modules |

# SMD Chip Sizes — Imperial Codes

0201 / 0402 / 0603 / 0805 / 1206 / 1210 / 2010 / 2512



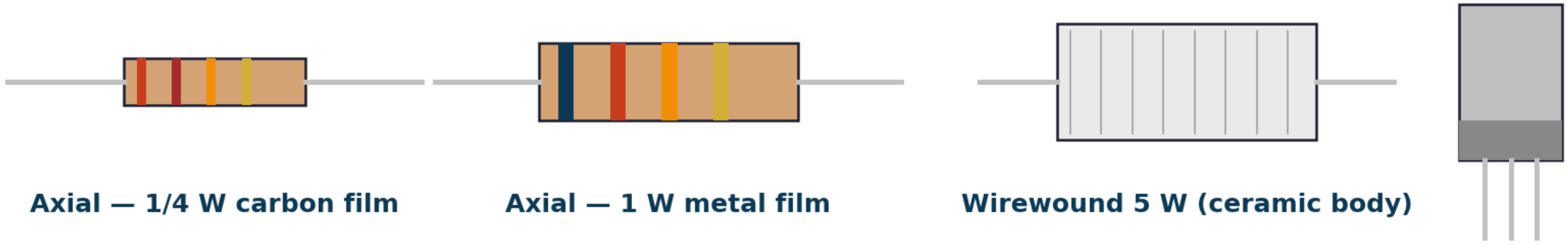

**How to read the code — e.g. '0805' = 0.08" × 0.05" = 2.0 × 1.25 mm**

Industrial baseline: 0603/0805. Mobile/wearable: 0402 → 0201. High-power: 1206 → 2512. Metric code exists too (e.g. 0805 = 2012M).

# Through-Hole Resistors

Axial bodies, color bands, power dissipation

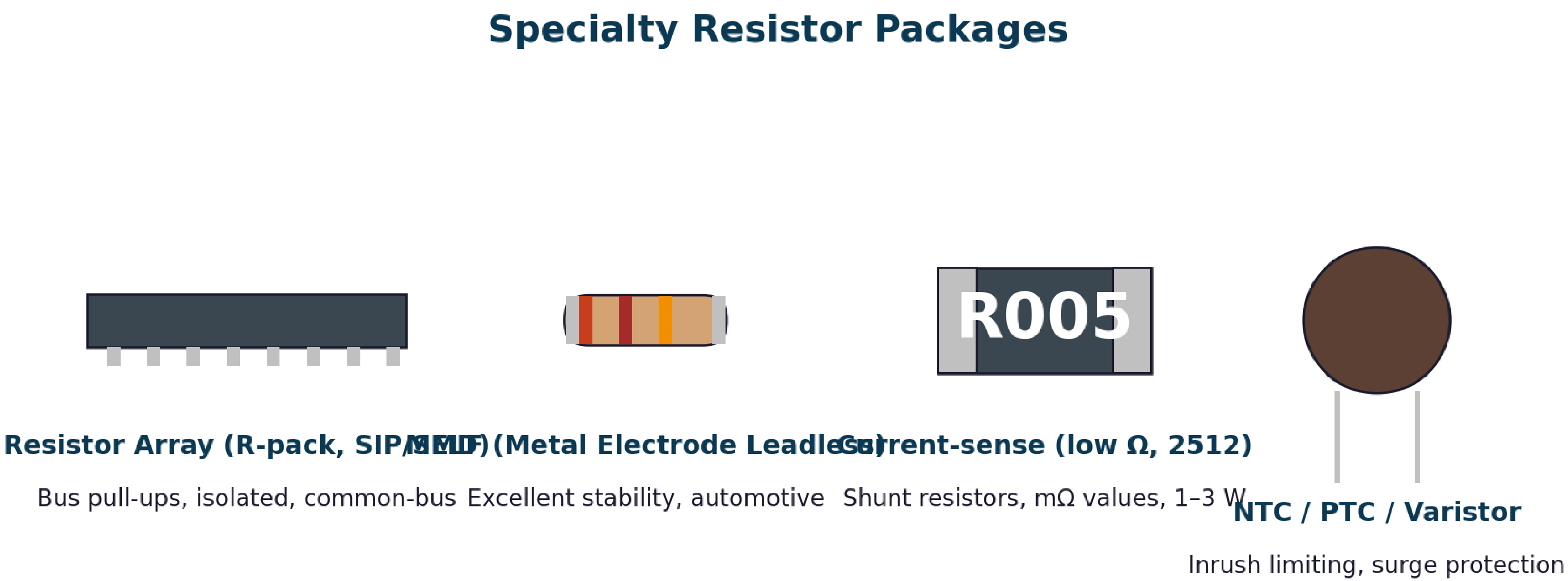
## Through-Hole Resistor Packages




*Standard E-series (E12 / E24 / E96) • Tolerances ±5% / ±1% / ±0.1% • TC: ±50 to ±100 ppm/°C*

# Specialty Resistor Packages

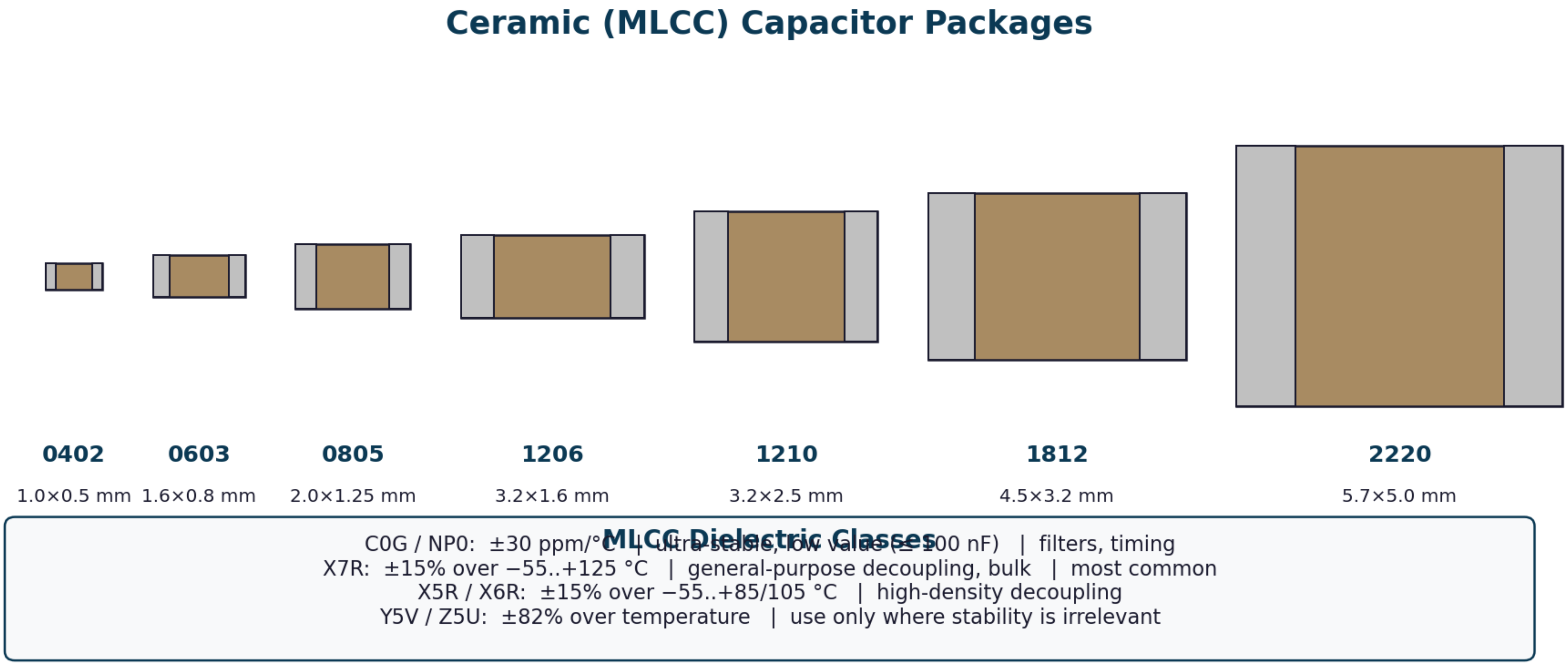
Arrays, MELF, current-sense, NTC/PTC, varistors




*Tip: pick MELF for high-stability analog, large SMD (2010/2512) for current sensing, and arrays to save BOM lines.*

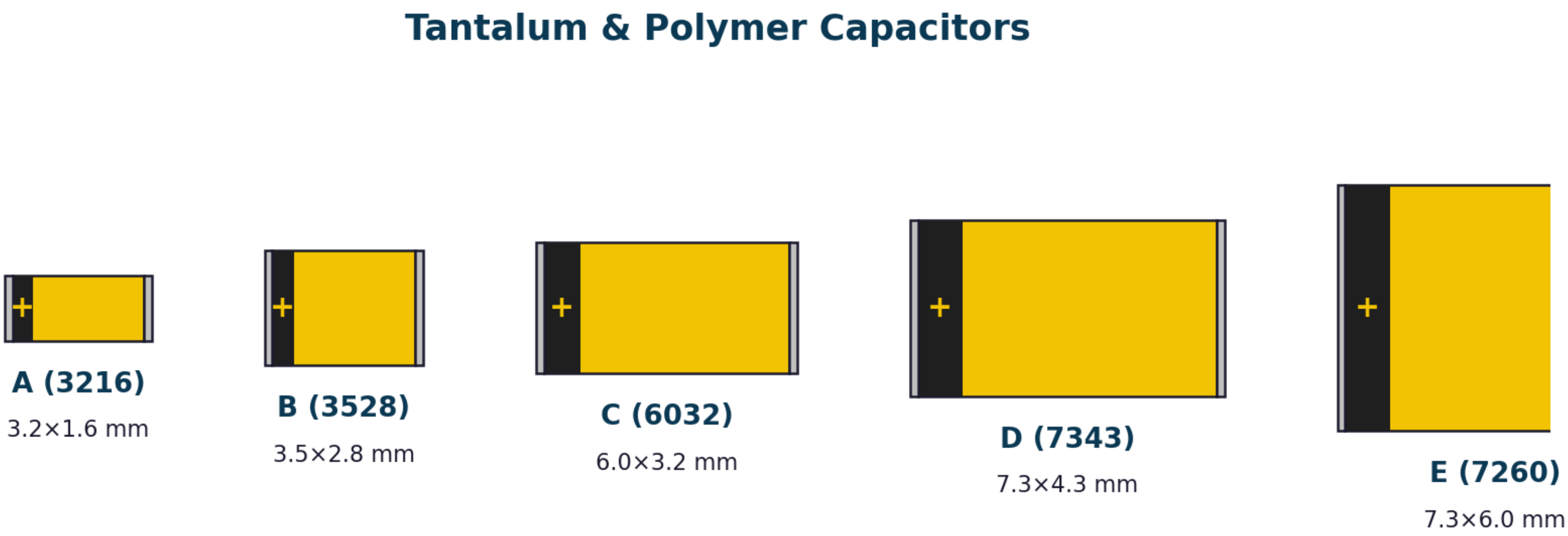
# Ceramic (MLCC) Capacitors

The workhorse — 0402 to 2220, C0G to Y5V



# Tantalum & Polymer Capacitors

Case codes A/B/C/D/E — polarised, low ESR

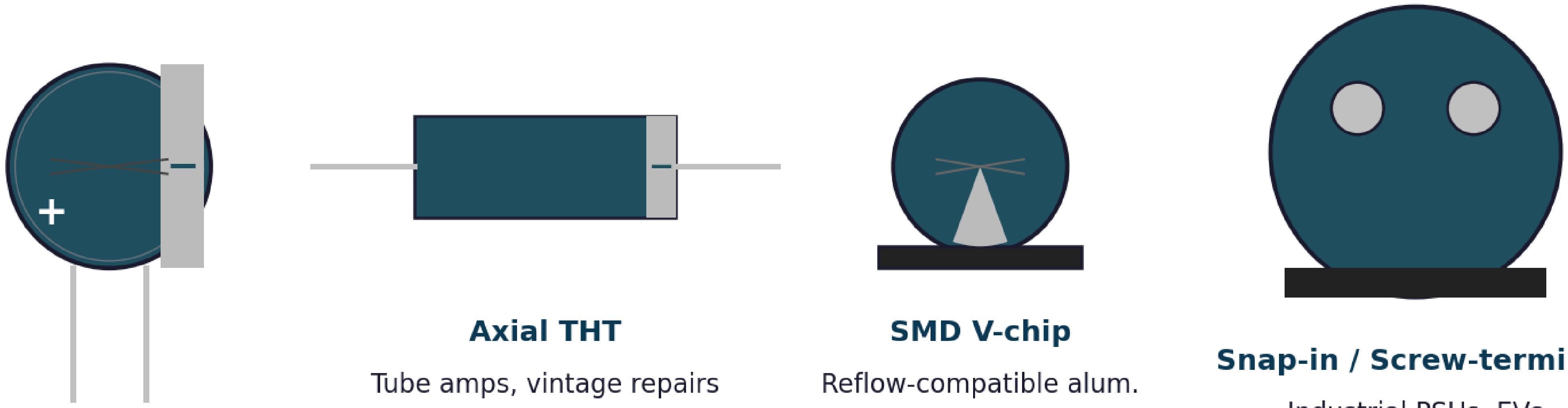



- Polarised — reverse voltage = failure (sometimes flames!)
- Choose 2× voltage derating; 3× for polymer tantalum
- Polymer Ta: lower ESR, safer benign failure mode
- Industry alternative to bulky aluminum electrolytics in dense designs

# Aluminium Electrolytic Capacitors

Radial / axial / SMD V-chip / snap-in

## Aluminium Electrolytic Capacitors




**Radial THT**
Up to 10 mF, 450 V

**Axial THT**
Tube amps, vintage repairs

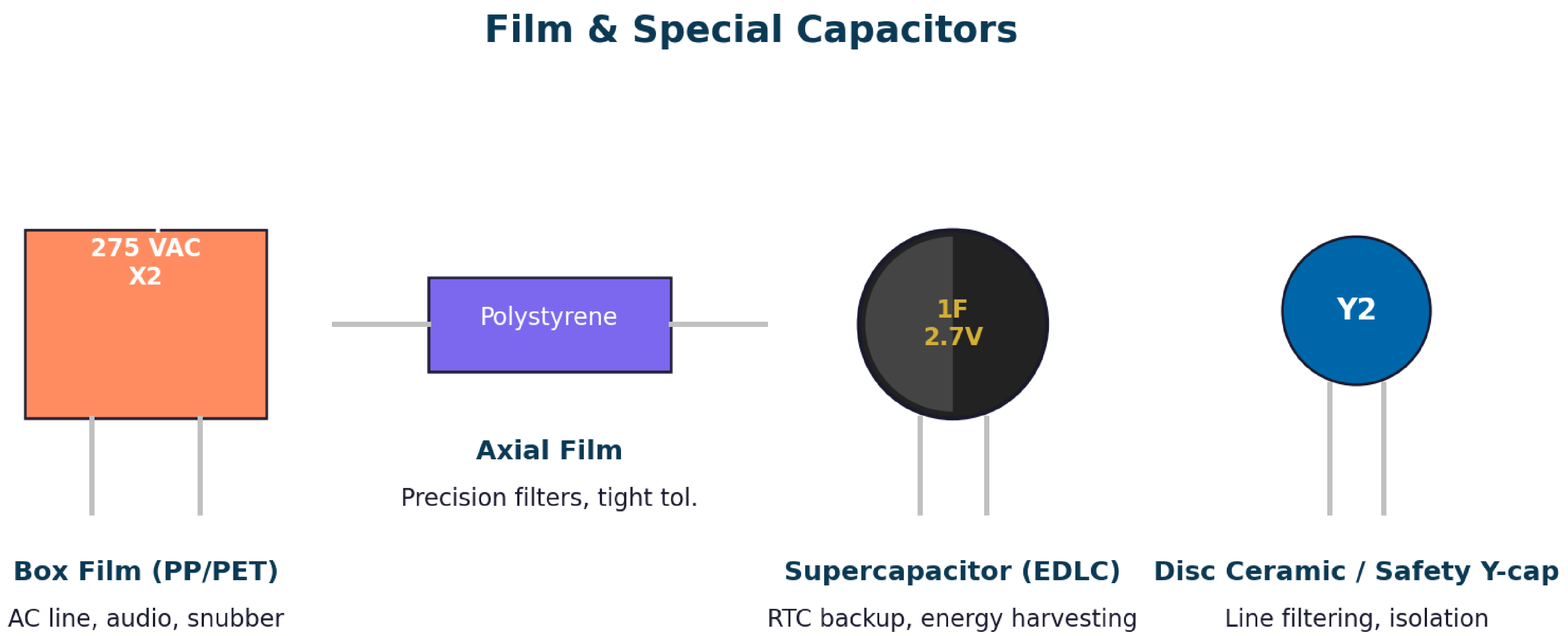
**SMD V-chip**
Reflow-compatible alum.

**Snap-in / Screw-terminal**
Industrial PSUs, EVs

*Polarised • Lifetime halves per +10 °C • Watch ripple-current rating, not just capacitance*

# Film & Special Capacitors

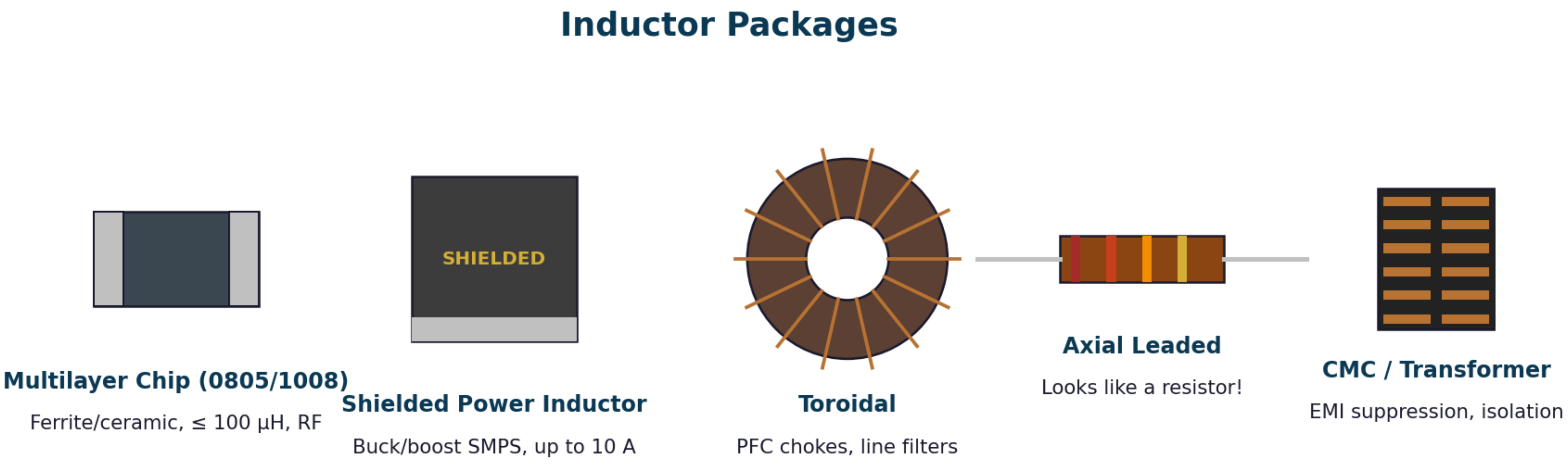
Polypropylene, polystyrene, supercaps, safety caps




*Film caps excel in linearity & lifetime; supercaps trade ESR for very large capacitance.*

# Inductor Packages

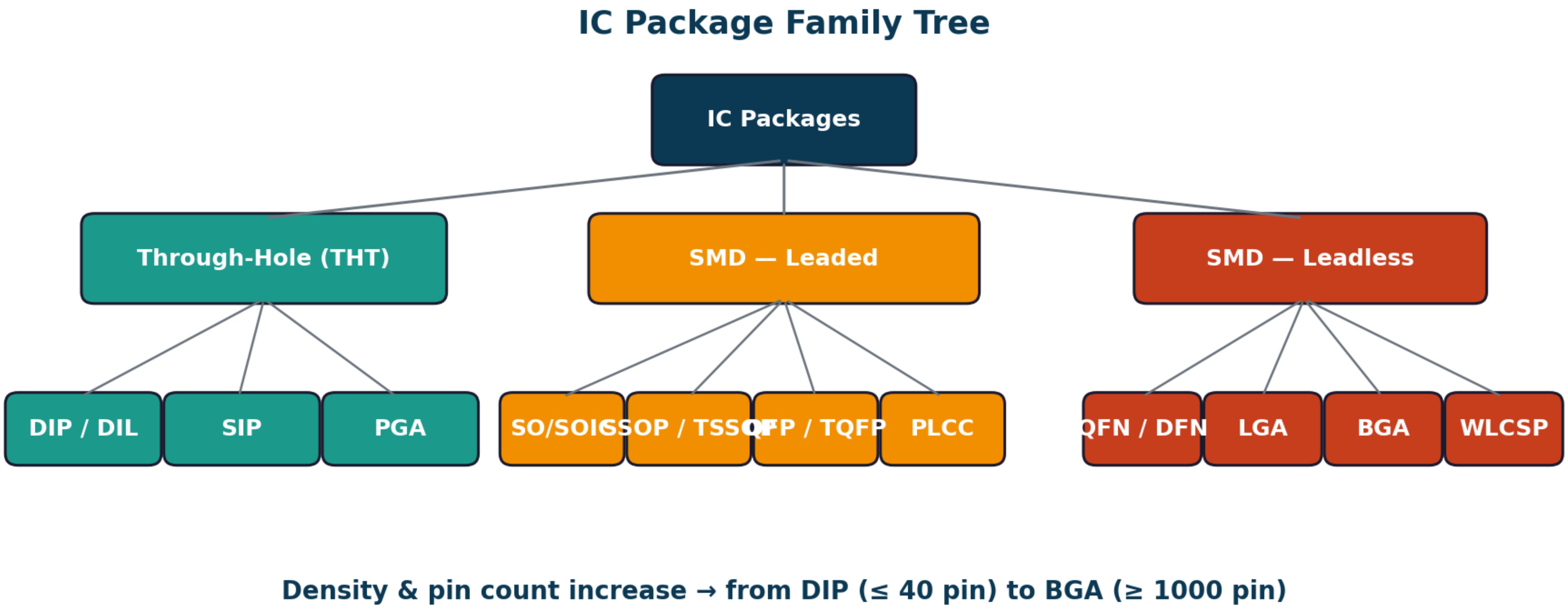
Chip, shielded power, toroidal, axial, common-mode chokes




*Shielded = magnetic field contained (low EMI). Unshielded = cheaper but radiates.*

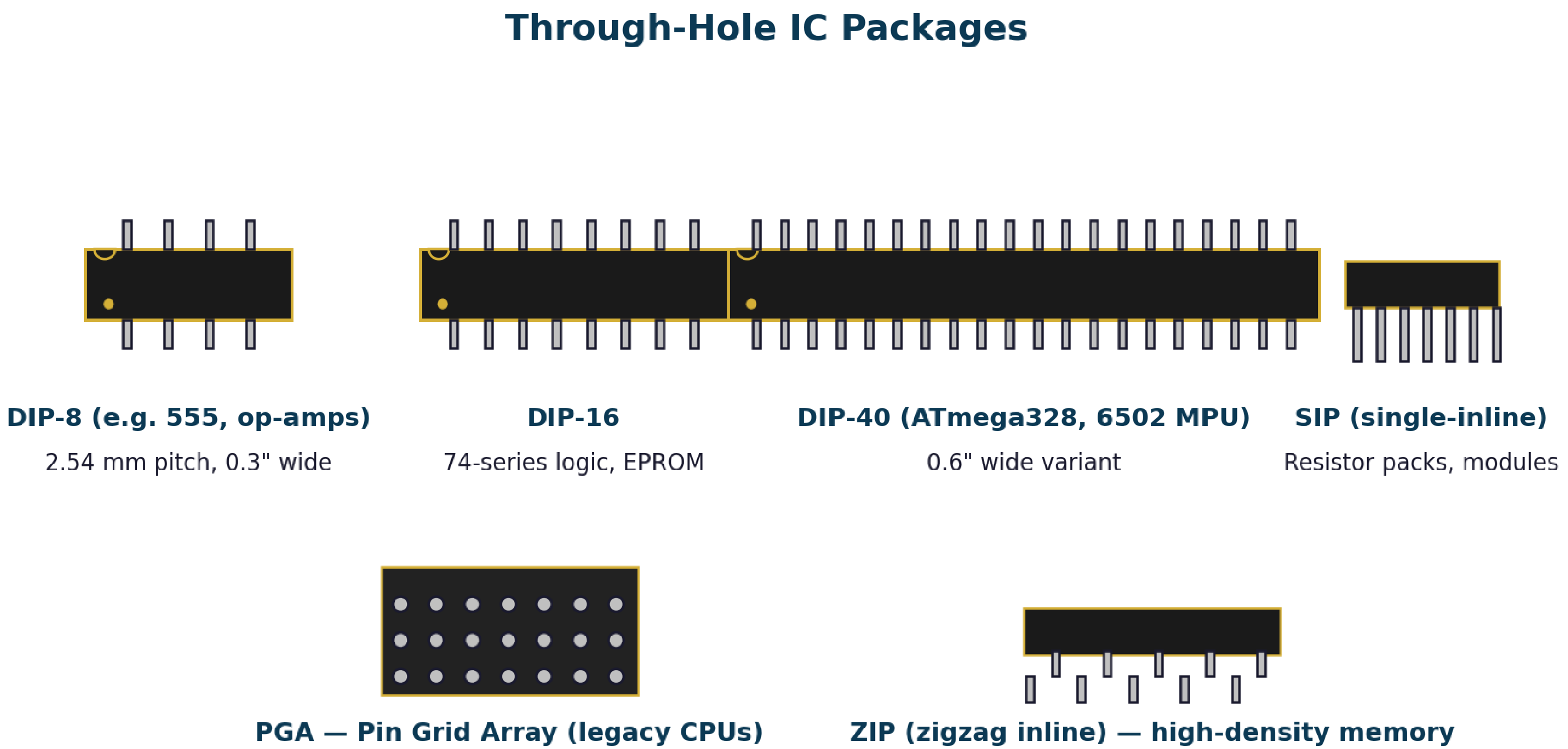
# IC Package Family Tree

Through-hole vs SMD-leaded vs SMD-leadless



# Through-Hole IC: DIP / SIP / PGA / ZIP

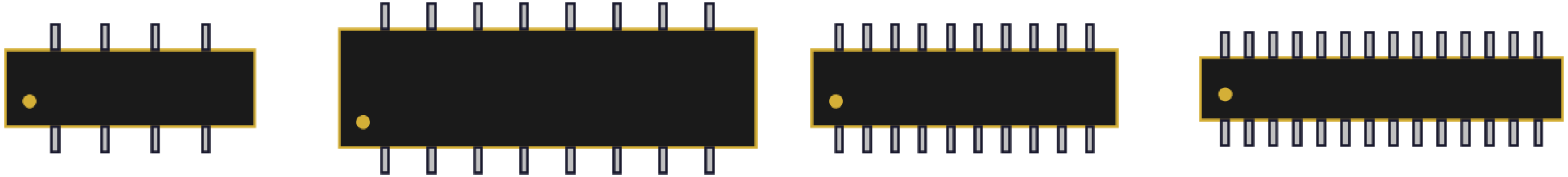
Where you'll still meet plastic dual-inline packages



# Small-Outline Family: SO / SOIC / SSOP / TSSOP

Gull-wing leaded SMD — easy to inspect, easy to rework

## Small-Outline (SO) Family — Gull-wing leaded SMD



**SOIC-8 (narrow)**
Op-amps, EEPROMs
**1.27 mm pitch**

**SOIC-16 (wide)**
Logic, interfaces
**1.27 mm pitch**

**SSOP-20**
Shrunk variant
**0.65 mm pitch**

**TSSOP-28**
Thin SSOP, MCUs
**0.65 mm pitch**

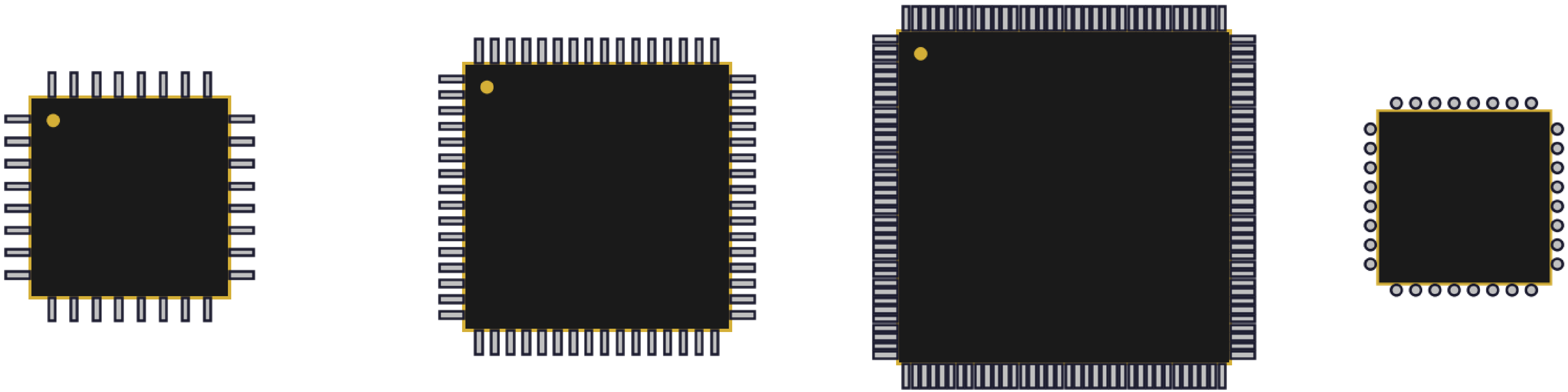
**Easiest SMD family to hand-solder**

MSOP (mini-SO), SOT-23 (3-lead transistor SMD), TSOP (thin SO for memory) — all share the same gull-wing concept

# Quad Flat Pack: QFP / TQFP / LQFP / PLCC

Gull-wing or J-leads on all four sides — pin counts 32 → 256+

## Quad Flat Pack (QFP) Family — leaded on 4 sides



**TQFP-32**
7×7 mm, 0.8 mm pitch
AVR / STM32 entry

**LQFP-64**
10×10 mm, 0.5 mm pitch
STM32F4 / ESP32 series

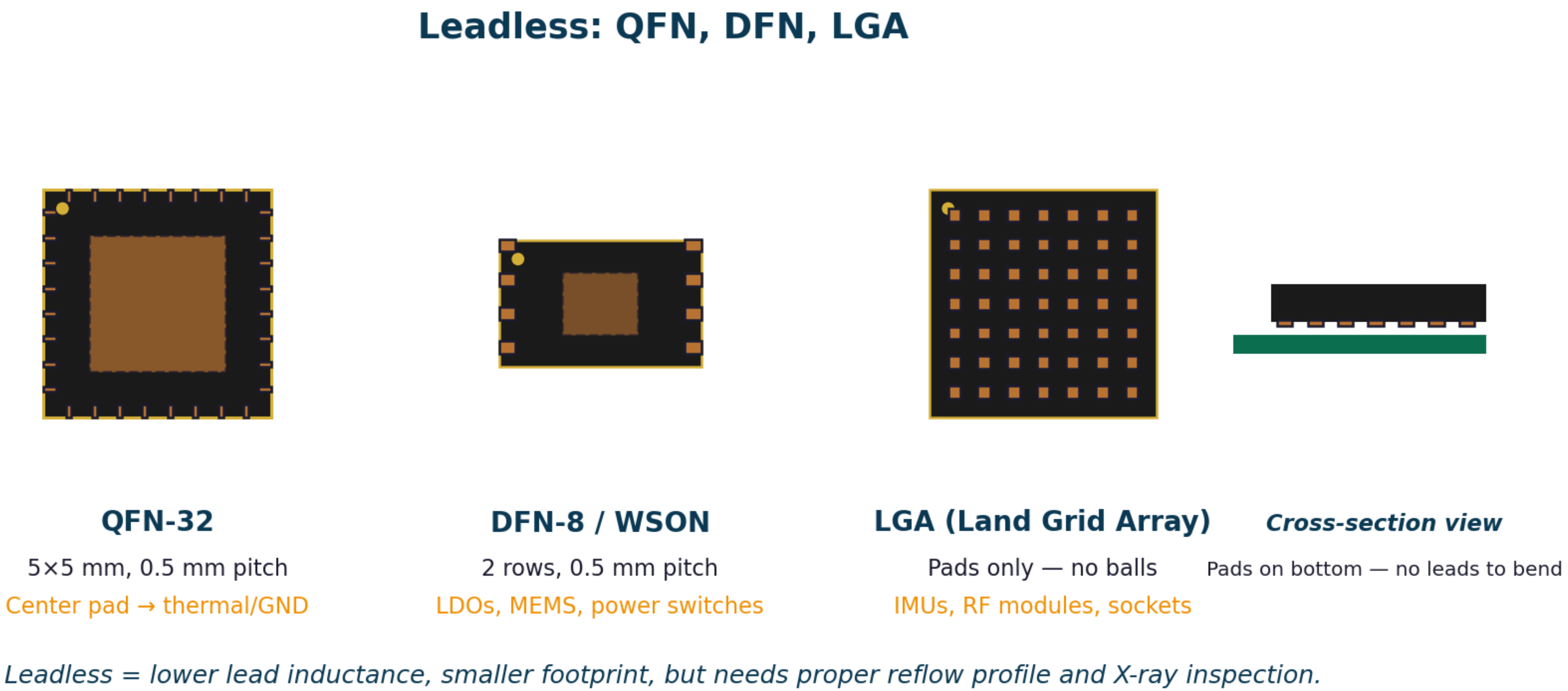
**LQFP-144**
20×20 mm, 0.5 mm pitch
High-end MCUs, FPGAs

**PLCC**
J-leads, socketable
Legacy EEPROMs, BIOS

*TQFP = thin, LQFP = low-profile. 0.5 mm pitch is the practical hand-solder limit.*

# Leadless: QFN, DFN, LGA

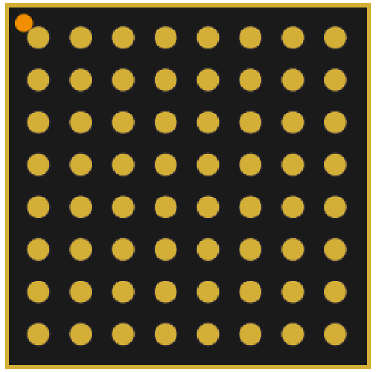
Pads underneath — modern default for power-conscious designs



# Ball Grid Array & Wafer-Level CSP

When pin counts cross 200 — and visible joints disappear

## Ball Grid Array (BGA) & Wafer-Level CSP



**BGA (top view)**

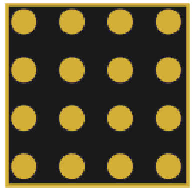
Solder balls instead of leads

1.0 / 0.8 / 0.5 / 0.4 mm pitch


**BGA cross-section**

Solder reflow forms joints

Inner balls = inaccessible (escape routing required)



**WLCSP**

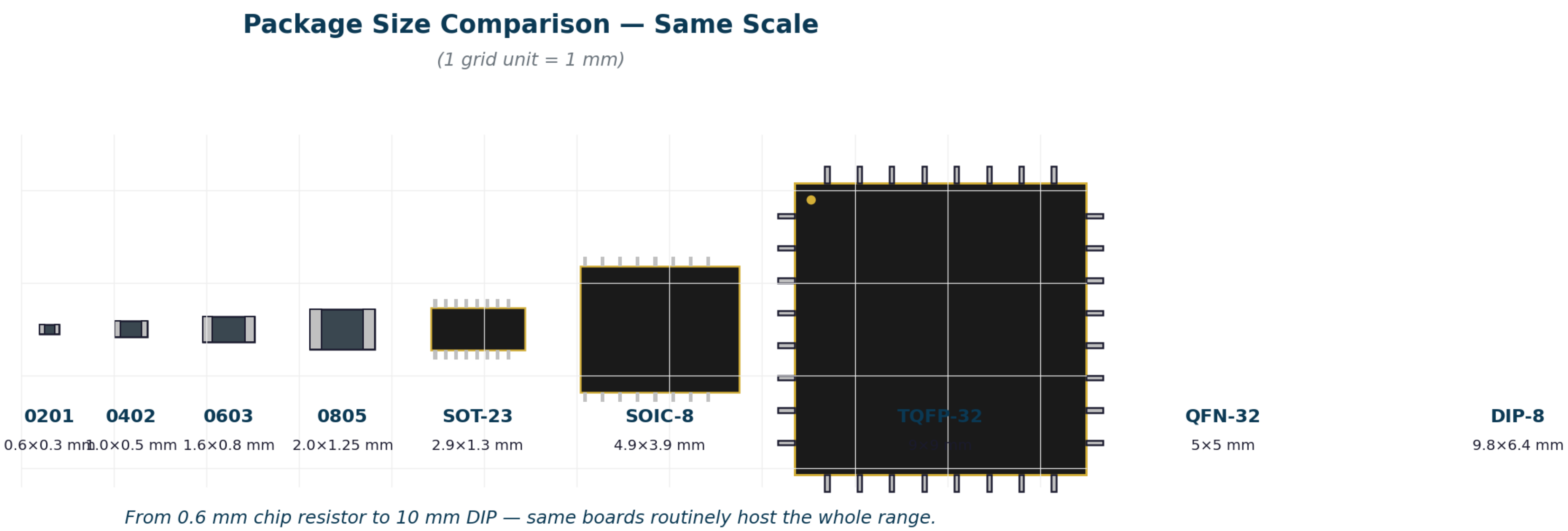
≤ 0.4 mm pitch, no substrate

Mobile SoCs, ultra-thin devices

*BGA = highest pin density (1000+ pins) — needs ≥ 4-layer PCB, X-ray inspection, no rework by hand.*

# Package Size Comparison — Same Scale

From 0201 chip to DIP-40 — drawn at 1 mm = 1 grid unit



# How to Choose a Package

Six criteria that drive the decision

## How to Choose a Package

### Pin Count

More pins → finer pitch / QFN / BGA

### Hand-Solder?

0603 / SOIC / TQFP friendly
QFN / BGA need reflow

### Board Density

Mobile = WLCSP/BGA;
Industrial = TQFP/QFN

### Thermal Load

Look for exposed pad / heatsink tab

### Cost & Lead-time

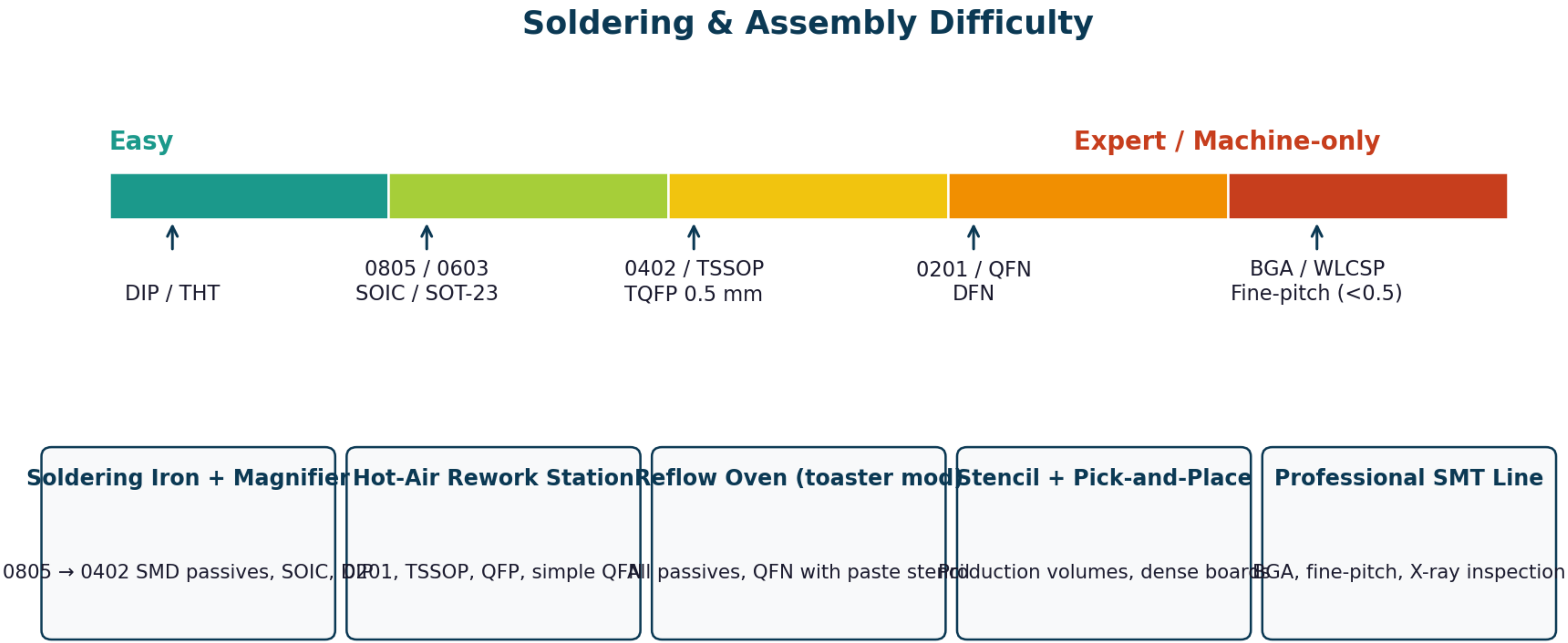
Smaller usually cheaper at volume;
larger easier in prototyping

### Reliability / Env.

Automotive / aero prefer leaded
(QFP) over BGA — visual inspection

# Soldering & Assembly Difficulty

What you can hand-build vs what needs machines



# Quick Reference Cheatsheet

Common packages, pitch, typical pin count, use

| Package | Pitch | Pins | Typical use | Hand-solder? |
|---|---|---|---|---|
| DIP-N | 2.54 mm | 4–40 | MCUs, op-amps, lab/proto | Easy |
| SOIC | 1.27 mm | 8–16 | Op-amps, EEPROM, drivers | Easy |
| SSOP / TSSOP | 0.65 mm | 8–48 | Logic, low-pin MCUs | Medium |
| TQFP / LQFP | 0.5 / 0.4 mm | 32–256 | MCUs (STM32, ESP32), FPGAs | Medium |
| QFN / DFN | 0.5 / 0.4 mm | 8–100 | LDOs, PMICs, RF, MEMS | Hard |
| BGA | 1.0–0.4 mm | 64–2000+ | SoCs, FPGAs, RAM, GPUs | Reflow only |
| WLCSP | ≤ 0.4 mm | 10–200 | Mobile PMICs, sensors | Reflow only |
| 0805 / 0603 | — | 2 | General passives | Easy |
| 0402 / 0201 | — | 2 | Dense / mobile passives | Hard / very hard |

# Industry Trends in Component Packaging

Where the next 5 years are heading

## Where Component Packaging is Heading

### 1 Miniaturisation

0201 → 01005 → 008004
passives shrinking faster than Moore

### 2 Integrated Passives

Multi-element arrays,
IPDs replacing 4 discretes

### 3 Chiplet / 2.5D / 3D

BGA modules with stacked dies
(HBM, Foveros, AMD chiplets)

### 4 Embedded Components

Resistors/caps inside PCB layers
→ freed surface area

### 5 Polymer & Hybrid Caps

Replacing aluminium electrolytics
— lower ESR, longer life

### 6 Lead-Free / RoHS / Halogen-free

Global compliance default

# Key Takeaways

Six things to remember from this lecture

### Package = Engineering Decision

Footprint, thermal, electrical, assembly, cost — all in one choice.

### Default to SMD

Modern boards are SMD-first.
THT only for connectors and heavy power.

### Start at 0603 for Passives

Step down to 0402 only if space demands.
Reserve 0201 for production.

### Choose Dielectric Wisely

X7R = decoupling. C0G/NP0 = timing/filters.
Never Y5V in a signal path.

### Leadless ≠ Lead-free

QFN / BGA hide their solder joints.
Use X-ray or trust your process.

### Read the Datasheet — Twice

Verify pitch, body size, thermal pad, and recommended landpattern.

# Questions & Discussion

Ask anything — package choice, soldering, sourcing, datasheets.

## References & Further Learning

- JEDEC.org — official package outline standards (JEP95, etc.)
- IPC-7351 — generic landpattern standard for SMD components
- Murata / TDK / Samsung MLCC selector tools — capacitance vs DC-bias data
- Texas Instruments — package summary tables in every datasheet
- Eric Bogatin — Signal & Power Integrity Simplified (package parasitics chapter)
- Phil's Lab (YouTube) — SMD soldering + datasheet reading walkthroughs
- Application notes from Vishay, Yageo, KEMET on derating and reliability
- Pick-and-place machine vendor application notes — practical assembly limits

Thank you • Build something this week — even a breakout board.